MODULARRAGOPTIMIZATION · GLOBAL CORRELATION EXPERIMENT

LLM Scores vs. Embedding Proximity to Ground Truth — Summary

Completed 2026-08-08 **Pairs** 1,000,174 **Queries** 199 (187 with GT) **Pool** 5,026 docs · fixed for all queries

Model gemini-2.5-flash-lite · NVIDIA IH

Hypothesis confirmed. LLM scores are consistently and significantly negatively correlated with cosine distance to ground-truth — across all 199 queries with no exceptions. The LLM outperforms the retriever as a GT identifier (99.5% vs. 73.8%) and fully recovers all retriever failures. Core limitation: score saturation at 1.0 ties GT with a median of 18 other docs per query, preventing single-document selection.

KEY NUMBERS AT A GLANCE

SPEARMAN R
−0.290
Median · 100% negative

PEARSON R
−0.503
100% negative · 100% sig.

ROC-AUC
0.997
Near-perfect GT detection

LLM RECOVERY
99.5%
186/187 queries

RETRIEVER
73.8%
138/187 found GT

TIES @ TOP
18
Median docs at score 1.0

COMPONENT ACCURACY — 2×2 (N = 187 QUERIES WITH GT)

	LLM RANKED GT #1	LLM MISSED
RETRIEVER FOUND GT	**137** 73.3% — Both correct ✓✓	**1** 0.5% — retriever only (query 29 only)
RETRIEVER MISSED GT	**49** 26.2% — LLM recovers all retriever failures ✓	**0** 0.0% — neither fails simultaneously

> **Critical insight:** The LLM fully compensates for all 49 retriever failures (26.2% of queries). Combined pipeline = 100% recall. Only 1 total failure across both components (query 29).

CORRELATION: LLM SCORE VS. DISTANCE-TO-GT

METRIC	MEDIAN	MEAN	% NEGATIVE	% SIGNIFICANT
Spearman r	**−0.2900**	−0.2832	100%	100%
Pearson r	**−0.5031**	−0.4937	100%	100%
Kendall τ-b	**−0.2322**	−0.2275	100%	100%
Point-biserial (vs. is_GT)	**+0.1188**	+0.1205	0%	99.5%

SCORE SATURATION & TIE ANALYSIS

METRIC	VALUE	NOTE
GT score = 1.0	**186/187 (99.5%)**	—
GT is unique top scorer	**0/187 (0.0%)**	Always tied with ≥1 other doc
Median docs tied at GT score	**18**	Range: 2–38
Queries with <10 tied docs	30/187 (16%)	Tightest discrimination cases
Queries with <20 tied docs	135/187 (72%)	Manageable with secondary ranker

TOP-5 SCORES VS. GT SCORE (PER QUERY)

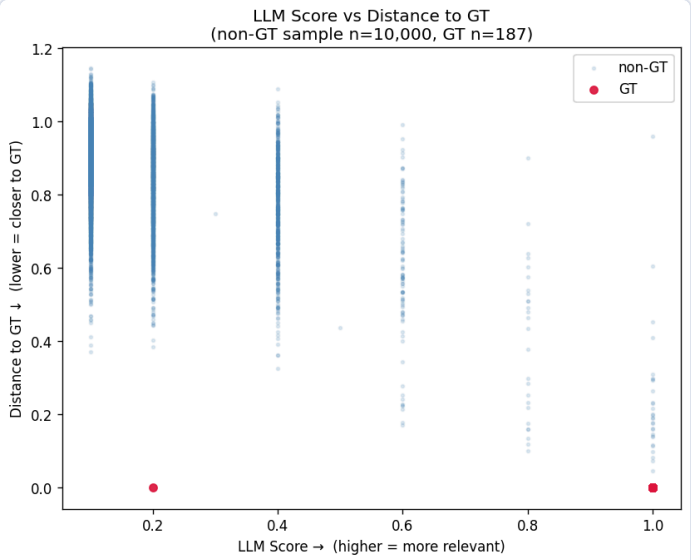
RANK	MEAN SCORE	TIED WITH GT	SCORE = 1.0
#1 (GT / co-rank)	1.000	**186/187 (99%)**	186/187
#2	1.000	**186/187 (99%)**	**187/187 (100%)**
#3	1.000	**186/187 (99%)**	**187/187 (100%)**
#4	0.999	185/187 (99%)	**186/187 (99%)**
#5	0.996	182/187 (97%)	183/187 (98%)

NON-GT SCORE DISTRIBUTION (999,987 DOCS)

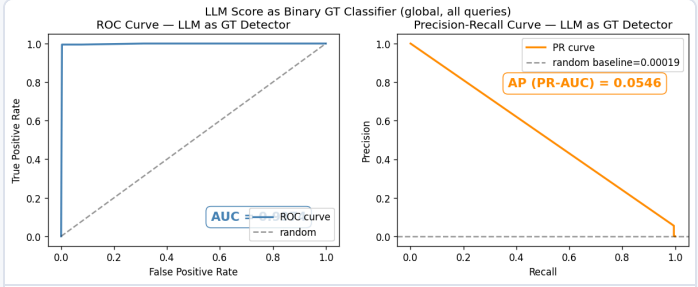


Score quantization: The model effectively uses 6 discrete levels. 92.3% of non-GT docs score ≤ 0.2. Only 0.32% (3,200 docs) reach 1.0 — these are the false-positive ties with GT. MRR optimistic = 0.995 vs. tie-corrected = 0.135.

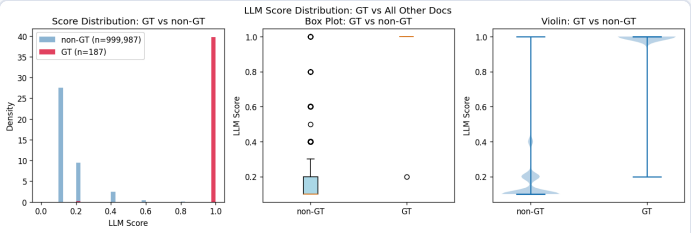
CHARTS



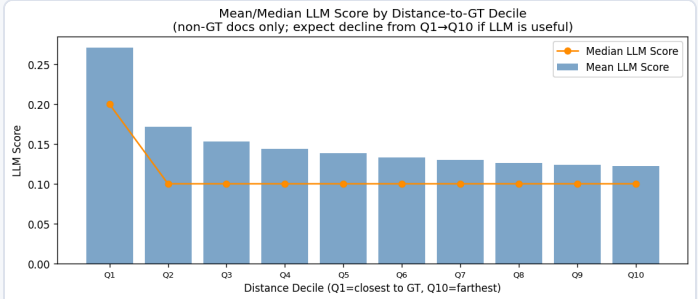
LLM score vs. dist_to_gt: GT cluster top-left (score 1.0, dist ≈ 0) vs. non-GT bulk lower-right.



ROC-AUC = 0.997. PR-AUC = 0.055, but 292× the random baseline of 0.000187.



GT mean = 0.996 vs. non-GT mean = 0.151. Near-complete separation.



Score by distance decile: 0.271 (closest) → 0.122 (farthest). Monotonic gradient.

Run: `staged_run_2026-08-03_08-32-45` · 1,000,174 pairs · gcp/google/gemini-2.5-flash-lite via NVIDIA Inference Hub · 2026-08-10